

CSM-322: Information and Coding Theory

Lab - 5

(Odd Semester 2024-25)

October 3, 2024

Q1. Linear Codes

Given a nonempty subset S of F_2^n . Write a program to find a generator matrix and a parity-check matrix for the binary linear code $C = \langle S \rangle$.

Example: $S = \{11101, 10110, 01011, 11010\}$

Output:

G =

```
1 0 0 0 1
0 1 0 1 1
0 0 1 1 1
```

H =

```
0 1 1 1 0
1 1 1 0 1
```

Q2. Linear Codes

Let G be a given generator matrix for a code C (in particular take binary linear code). Write a program to print the parity-check matrix for the given generator matrix.

Q3. Linear Codes

Let H be a given parity-check matrix for a code C (in particular take binary linear code). Write a program to print the generator matrix for the given parity-check matrix.